Verifying the Theoretical Predictions of Fourier optics on a Four-f system

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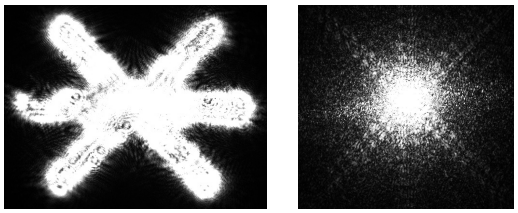
L2 Research Led Investigation, Fourier Optics, Lab day: Thurs

Abstract

This report details an experiment to verify the description of the output of a four-f system with a diffraction grating placed in the Fourier plane by a convolution of the input function and the Fourier transform of a comb function associated with the grating. The two relationships being analysed were the intensity within overlap of image copies as a function of horizontal diffraction grating displacement and the image copy separation in the output plane as a function of wavelength of light, lens focal length and grating slit separation. The first relationship fitted well, showing a good fit while the second relationship fitted with a reduced χ^2 value of 0.72 when a lens of 50cm focal length was used and of 0.08 when a lens of 20cm focal length was used. These results show that our descriptions of the output of the system used were accurate. This does not show how other functions affect the output of the same system which could be an area of further study to verify a more general prediction of this type of system.

1 Introduction

Fourier optics provides a detail of the propagation of light through a variety of apertures, lenses and screens. This is clearly an important aspect of photography and optical imaging where lenses are used to focus light. The input and output functions of a single lens system can be related by a Fourier transform; a case of a Fourier series when the period is taken to be infinite. Fourier series are used to represent periodic functions as a sum of simpler periodic functions, more specifically sinusoidal functions. Figure 1 shows an input function of an asterisk shape and the Fourier transform after this function has passed through a lens. The spatial frequencies have been separated out with the low spatial frequencies in the centre and high spatial frequencies further out. When dealing with lenses, we are concerned with Fraunhofer diffraction since the setup can also be replicated without the lens but in the far-field. [1] The Fourier transform returns a distribution of spatial frequencies on the Fourier plane which can be manipulated with the use of spatial filters to alter the output function. The mathematical effect of placing any kind of aperture function in the Fourier plane is to multiply the Fourier transform of the input function with the aperture function.



(a) (b)

Figure 1: The image of an asterisk shape (a) used throughout this experiment as the input function and after it has had a Fourier transform applied (b), taken at the Fourier plane of a single lens system

The four-f system is used as a basic setup for the analysis of Fourier optics and spatial filtering. It is comprised of a beam source, directed onto an aperture, which passes through a lens to be focussed onto the fourier plane, then passed through a second lens so that the output beam is collimated at the output plane. In a four-f system, the operations applied to the input functions are an initial Fourier transform as it passes through the primary lens, this is then multiplied by the aperture function placed on the fourier plane and a final Fourier transform is applied to this function at the secondary lens before being detected at the output plane. The final operation is a convolution of the input function, which has been flipped in the x- and y-directions, with the Fourier transform of the aperture function placed on the Fourier plane. One application of Fourier optics is in optical information processing. While currently the vast majority of information is processed digitally due to the flexibility and power of digital computers, the processing of information using optical systems is a growing area of innovation. [2]

A comb function is expressed as a series of Dirac delta functions separated by a constant length, shown in Equation 1.

$$\text{Ш}_T(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT) \tag{1}$$

where T is the period. The Fourier transform of the comb function also takes the form of a comb function, which can be shown mathematically to provide a repeating pattern of the function it is convoluted with. This means that if a comb function is placed in the Fourier plane of a four-f system, the output is a repeating pattern of the input function, where the overlap between repeating units either interferes destructively or constructively depending upon the horizontal position of the comb.[3] The relationship between the horizontal displacement, x' of the diffraction grating and the intensity of light, I, in these overlaps can be shown to follow the relationship shown in Equation 2, where d is the slit spacing of the diffraction grating. This relationship is derived from a convolution of the input function and the Fourier transform of the diffraction grating's comb function.

$$I \propto cos^2 \frac{\pi x'}{d} \tag{2}$$

The intensity relation at the output is shown in Equation 2

which shows the intensity is only dependent on the grating slit separation and grating position, not the wavelength of light or lens focal length. This convolution can also be used to show that the separation of the image copies at the output plane is given by Equation 3, where λ is the wavelength of light, f is the focal length of the lenses and d is the grating slit separation.

$$x = \frac{\lambda f}{d} \tag{3}$$

A comb function can be described by a diffraction grating which is only an approximation since the slits have a definite width and a Dirac delta function is impossible practically.

2 Methods

A four-f system was used for this experiment where a collimated laser beam was input and a monochromatic CMOS camera was used to record images of the output as a function of intensity, shown in Figure 2. These images were imported to Microsoft Paint software to take positional measurements which was preferred as it gave a cursor position in terms of pixels. All elements were attached to an optical breadboard which allowed us to increase the precision of the experiment. Care was taken to ensure that the beam travelled along to optical axis of the system so that the centre of all apertures was illuminated to provide the maximum definition at the output plane. The beam source used had a setting to vary the wavelength of light to set values which allowed us to keep the beam source consistent.

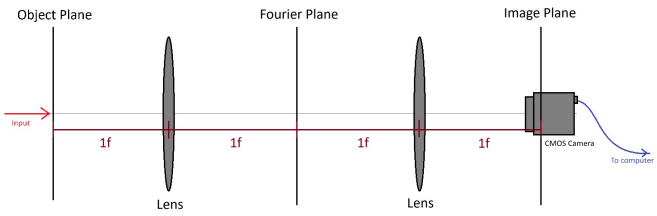


Figure 2: The experimental setup of the four-f system where the input light along the optical axis comes from a collimated laser source and 1f refers to the length of one focal length of the lenses. For this experiment, a diffraction grating was placed on the Fourier plane

A beam expander was used to collimate the beam and increase the diameter of the beam so that the input function could be fully illuminated. Measurements of the properties of the components of the four-f system were taken first. The focal length of the spherical lenses used was determined by directing the collimated light through the lens and measuring the distance from the lens plane where the radius of the beam was at a minimum along the optical axis. The rest of the apparatus was spaced out by this measured length with a metre rule of 1mm precision. The wavelength of the light was given by the function selector on the beam source and the slit spacing was measured by placing the grating on the object plane and measuring the fringe separation at the output. The focus of this experiment was to observe the effects of functions applied in the Fourier plane of a four-f system on the output function. Primarily, we were verifying the relationship between the horizontal displacement of a diffraction grating at the Fourier plane and the intensity of light in the overlap regions of repeat units of the input function at the output plane. The diffraction grating was placed on a travelling stage at the Fourier plane so that it could be displaced by increments of 0.01mm and an image of the output was taken. The images were analysed to measure how the intensity of light in certain regions varied. A python script was used to extract the average intensity readings within these regions. This could be plotted against grating displacement and a χ^2 curve fit was applied using the model function. The best fit model parameters could then be compared to the expected values to further verify the relationship. The other relationship we were verifying was the image copy separation at the output plane of the same experimental setup. Images at the output plane were taken for various different wavelengths of light, lens focal lengths and diffraction grating slit separation. These were analysed to determine the separation of the image copies.

3 Results

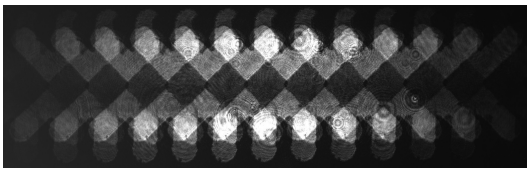


Figure 3: The image taken at the output plane when a diffraction grating of slit width 0.241mm is placed in the Fourier plane of a four-f system of focal length 20cm when using a laser beam of wavelength 940nm and an asterisk shape is placed in the object plane

Grating number	Measured slit spacing (μm)	error (μm)
1	242.9	0.4
2	241.5	0.4
3	125.2	0.1

Figure 4: The measured values for the slit separation of each diffraction grating when placed in the object plane of the four-f system and measured at the object plane

The values for the slit separation in Figure 4 were measured to provide a comparison to the values obtained from the translational data seen in Figure 4 to provide further verification of the expected relationship.

Grating number	Measured slit spacing (μm)	Difference (%)
1	252	4
2	245	1
3	127	1

Figure 5: The values for the slit spacing, calculated from the period of equation 2 fitted function to the data of Intensity against grating displacement and the percentage difference from the values in Figure 4

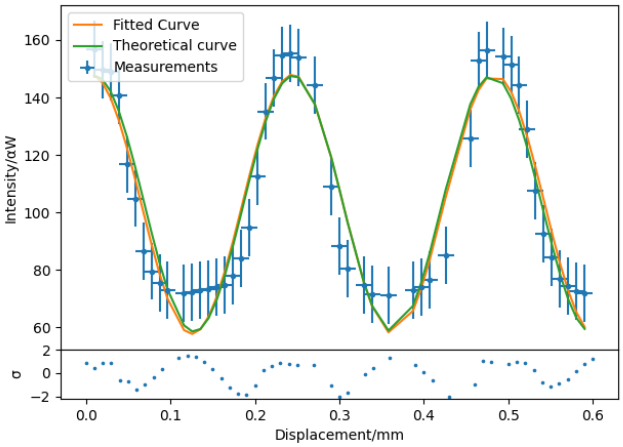


Figure 6: The average intensity recorded for a region on the CMOS camera against the displacement of the Diffraction grating at the Fourier plane for Coarse Grating 1, Lens 2, and an incident wavelength of 940nm with normalised residuals shown underneath. α is an arbitrary constant of proportionality between the intensity of light at the image plane and the output of the CMOS camera. The y-error bars shown have length of the standard error of the residuals and the x-error bars have length 0.01mm

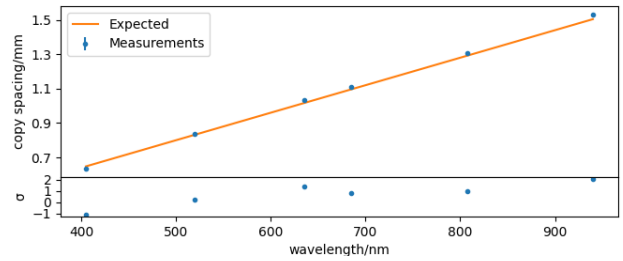


Figure 7: The plot of measured image copy separation against wavelength for lens 2 and diffraction grating 3 with the measured data and the expected values based on Equation 3 with normalised residuals shown underneath. Error bars are too small to be seen.

Lens	reduced χ^2
1	3.9
2	2.3

Figure 8: The reduced χ^2 values returned for each lens when comparing the data for the image copy spacings to the relationship shown in Equation 3.

4 Discussion

Figure 3 shows the output of the four-f system for diffraction grating 2, lens 2 and a wavelength of 940nm. This was the expected output, where there is a repeating pattern of the input function used, in this case an asterisk shape, and the intensity of the pattern as a whole is highest in the middle, along the optical axis, and gets lower further away due to the finite width of the beam at the Fourier plane. There are regions of four definite intensities; of zero for where there are no images, the first intensity where there is no overlap but one image copy, regions where there is the overlap of two repeat units and the regions of greatest intensity are three units overlapping. This pattern shows constructive interference because the overlap regions are at a greater intensity. Depending on the properties of the setup, regions could exist where more than two units overlap due to a small image copy separation. This would increase the maximum intensity measured but not the period of oscillation of the intensity so would not affect the results. Therefore a large area could be used to get an average intensity which would have reduced the sources of random error in the measurements. Some datasets displayed a secondary, minor peak in intensity between the two maxima, which occurred when the region being analysed was either side of the central stripe which can be seen in Figure 3. Another cause of this flattening was that the maximum was moving across the overlap regions of some datasets as the grating was translated. This could have been due to the finite width of the slits which don't represent perfect Dirac delta functions. The two curves seen in Figure 6 are the theoretical curve, shown in green, which used the period derived from directly measuring the slit spacings to calculate the period of oscillation of the relationship and fitted to the arbitrary amplitude of the measured data and the fitted curve, in orange, which shows the curve when the period of the relationship is fitted to the data. By inspection, these two curves both show clear similarity both to the measured data and to each other. Figure 5 shows the values for the slit spacing calculated from the period of the fitted curve shown in Figure 6. The low percentage deviation from the values obtained by measuring the spacings directly also demonstrates that the model fits the data well. These two pieces of evidence led us to believe that the model in Equation (2) accurately described the data we obtained. The normalised residuals shown show some structure which was due to some of the pixels becoming either saturated or returning a zero value before the minimum value was reached especially at the minima which shows a large number of pixels reaching a zero value before the true minimum was reached. This was a larger issue with the datasets where more than one image copy occupied the region being analysed since the amplitude of the intensity oscillation was greater so the limits of the CMOS camera were more readily reached. This was shown in some figures where the extrema were flattened after reaching a maximum or minimum value. This didn't affect the value obtained for the grating slit separation as it did not affect the period of the fitted curve. Another consideration is that an Airy pattern was seen in the beam emitted due to an aperture within the beam source. This would not have had a large effect on the measurements but there would be small variations in intensity at the image plane due to this and would have been mitigated when taking an average over a region. Similarly, the finite width of the grating would affect the output intensity since it behaves as a multiple slit rather than a true diffraction grating. In our particular setup, we used mirrors to keep the path of the beam within the breadboard which may have added some distortion to the beam. The variations in the setup induced when setting up the system between lab sessions and swapping out the lenses

would have altered the sharpness of the image at the output so measurements for a single lens setup were all taken together where possible. It can be seen that the vertical position of the curves is skewed by the fact that the minima have been flattened while the maxima have not been. This was ignored because it didn't have a large effect on the measurement of the period. Some error was also seen due to the finite number of cycles that were recorded. More measurements within each dataset would reduce this error but we were limited for time and took as many as was possible. If more time was allocated, more datasets could have been measured using more gratings. For all datasets, a systematic error for the period was found where the fitted curve has a larger period than the theoretical curve, leading to a value for the slit spacing being smaller than the measured value. This could have also been a source of systematic error when measuring the slit spacings directly. There were 11 different datasets which all returned nearly identical curves for fitted and theoretical relationships. Each dataset contained a different combination of light wavelength, grating and lens focal length. The introduction of more variables helps to ensure the independence of the data from these variables. A value for the error in the intensity measurements could not be determined for a χ^2 goodness of fit test because the quality of data obtained relied mostly on the region chosen to analyse and the properties of the system which we were not able to express quantitatively.

When measuring the image copy spacings, the maximum number of repeat units was included to reduce the error in this measurement. The nature of some images and the finite width of the CMOS camera sensor meant the error was lower because there were more units displayed to measure across. For some images taken, there was not a definite edge of repeat units to take measurements from because the intensity became low for repeat units further away from the optical axis so a trade-off needed to be made to achieve the lowest error. In addition to this, some images had the image copies close together, making it difficult to differentiate between them. The grating for some images taken was not perfectly parallel to the optical sensor of the camera and the pattern was slightly skewed so this was corrected for when taking measurements. The image copy spacing returned a reduced χ^2 value of 3.9 for lens 1, shown in Figure 8 and 2.3 for lens 2 when using the relationship shown in Equation 3. The high values for reduced χ^2 suggests that the error used in calculating the reducedχ^2 was an underestimation of the actual error for this data. This can be seen in Figure 7 because the error bars are too small to be seen while, by inspection, the straight line fits the data well. The data from the two lenses was given separately due to the variations in the setup which would have occurred and led to two different reduced χ^2 values. The pixel width was found on the manufacturer's website for the CMOS camera which allowed us to convert the measurements in pixels to metres. This value was 5.86μm which was taken to be exact.

5 Conclusion

These experiments have shown that the relationships derived from the convolution of the Fourier transform of the grating function at the Fourier plane with the input function can accurately predict the output function of a four-f system. Our initial observations were as expected and our data returned a value of grating slit separation within 4% of the values obtained from measuring the slit separation directly and a minimum χ^2 value of 2.3 and 3.9 for the measurements of the image copy separation as a function of light wavelength, lens focal length and grating slit separation. This shows that we must accept the null hypothesis that the output of a four-f system can be described by the convolution of the input function with the Fourier transform of a comb function when a diffraction grating is placed on the Fourier plane. Further study could be to verify the same approach to predicting the output of this type of system but for different functions applied in the Fourier plane to give a more general description of Fourier optics.

References

[1] E.G. Steward, Fourier Optics: an Introduction (Halsted Press, 1983)

[2] H.M. Ozaktas and M.A. Kutay, "Optical Information Processing: A Historical Overview", Digital Signal Processing **119**, 103248 (2021). doi: 10.1016/j.dsp.2021.103248.

[3] C.S. Adams and I.G. Hughes, Optics f2f: From Fourier to Fresnel (Oxford University Press, 2019), pp.159-176. doi:10.1093/oso/9780198786788.001.0001

[4] I.G. Hughes and T. Hase, Measurements and their Uncertainties : A Practical Guide to Modern Error Analysis (Oxford University Press, 2010), ProQuest Ebook Central, https://ebookcentral.proquest.com/lib/durham/detail.action?docID=584562.

Error Appendix

Grating translation data: The error in the displacement of the grating was taken to be 0.01mm due to the precision of the digital calipers used. The error for the measurements for setting up the apparatus were taken to be 1.4mm using calculus-based error propagation. The main source of error was from the image analysis. The standard error of the residuals, shown in Figure 6, was used for the error in the measurements of intensity which varied greatly for different datasets and are shown in Figures 9 and 10.

	λ (nm)	
Grating	520	635
1	7.9	2.9
2	5.2	2.6
3	6.1	2.9

Figure 9: The standard errors for the measurements of intensity for lens 1. Units of αW

	λ (nm)	
Grating	635	940
1		2.7
2	37.1	7.8
3	23.5	11.3

Figure 10: The standard errors for the measurements of intensity for lens 2. Units of αW

Image copy spacing data: The wavelength was taken to be exact from the beam source. The error of the focal length of the lens was taken to be 1cm since we measured the value and was within 1cm of the value ststed for the lens. The error on the slit spacing was derived from the grating translation data. The errors in the lens focal length, grating slit separation and wavlelength of light were calculated from the extreme value of χ^2 contour parallel to the axis of that variable. Each variable's error is shown in Figure 11 which agree with the assumed error in the apparatus.[4]

value	error
Coarse grating 1	0.11mm
Coarse grating 2	0.06mm
lens focal length	1cm

Figure 11: The errors in variables calculated from the $\chi^2 + 1$ contour

The values from Figure 11 are derived from the χ^2 + 1 contour of the three dimensional χ^2 space so a contour plot was not able to be included.

Scientific Summary for a General Audience

Lenses can be found everywhere from cameras to reading glasses and Fourier optics is the study of light as it passes through a lens. Mathematical equations are able to describe what will happen to a beam of light when it is directed onto a lens. When a beam where all the light is travelling parallel so that the beam looks like a cylinder, say, is shone onto a lens, it changes the cylinder to a cone where the focus point is the tip of the cone. The light will also be separated out into different components at this point and we can remove or alter parts of it. Then we are able to get this light back to look like a cylinder with the use of another lens so we are able to see what effect was made by removing parts of the beam. This is important because if we are able to know what is causing certain effects in cameras which produces a worse quality image, we will be able to alter certain components to reduce these effects.